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Two novel high-porosity materials as volumetric receivers for concentrated solar radiation

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Abstract

This article reports on two novel porous materials, which have been foreseen as volumetric receivers for concentrated solar radiation: a double-layer silicon carbide foam and a screen-printed porous silicon carbide material. Volumetric receivers are used in the solar tower technology. In this technology ambient air flows through the porous solid, which is heated by concentrated solar radiation. A heat exchanger then transfers the energy to a conventional steam turbine process. The general thermophysical and permeability properties of materials required for this application are reviewed. Experimental set-up and results of pressure loss and laboratory scale tests in concentrated solar radiation are reported. For the foam, efficiency data could be determined from the test results. Finally, a comparison is presented between the efficiency properties of the foam and those of materials used for the same application until now. This comparison shows, that the efficiency of the double-layer foam material is significantly higher. Up to now, porous materials consisting of a parallel channel geometry with thin walls showed disadvantageous permeability properties. By applying a new manufacturing process and modifying the channel geometry, the permeability properties of the printed material could be significantly changed, so that it now meets the requirements for an application as a volumetric receiver.

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Keywords: Volumetric receiver; Solar thermal concentrating technologies; Silicon carbide; Ceramic foam

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1. Introduction

Volumetric receivers consist of high-porosity materials and are used in the solar tower technology for the conversion of concentrated solar radiation into heat. This heat is then used to feed a conventional steam turbine process. As a heat transfer medium ambient air is used, which flows through the open pores of the material. This technology has been developed during the past 20 years, a number of large scale tests have been carried out [1,2]. A first 10 MW_{el} commercial plant is currently planned for Southern Spain [3,4]. The principle of the receiver is illustrated in Fig. 1, along with a typical flow chart of a solar power plant in Fig. 2. The general requirements and the resulting material properties of a material to be applied as a volumetric receiver are summarized in Table 1.

A lot of theoretical research work has been conducted on properties and physical restrictions of the open volumetric receiver principle [5,6]. The results significantly depend on how detailed the numerical calculation is performed and how reliable the material properties assumed correspond to realistic material data. In general, due to the increasing viscosity of air with increasing temperature, the permeability of the hot channels of the receiver is lower. This leads to a mass-flow distribution shown in the right sketch of Fig. 1. In this case a Gaussian distribution of the incoming concentrated radiation is assumed.

It has been shown by Pitz-Paal and Hoffschmidt [7] and Buck [8], that the flow through the porous material is significantly influenced by thermal conductivity and permeability properties of the material. Obviously, using a long receiver made out of a material with a high thermal conductivity leads to a heat flow perpendicular to the

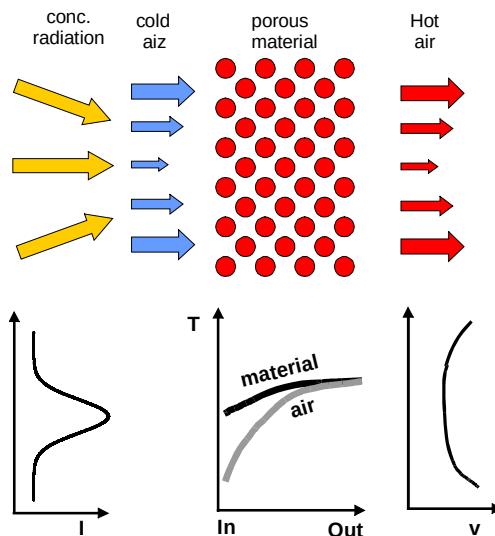


Fig. 1. Open volumetric receiver principle: Typical distribution of the incoming solar radiation across the aperture (left), material and air temperature distribution perpendicular to the aperture and along the direction of air flow (middle) and resulting mass flow distribution (right).

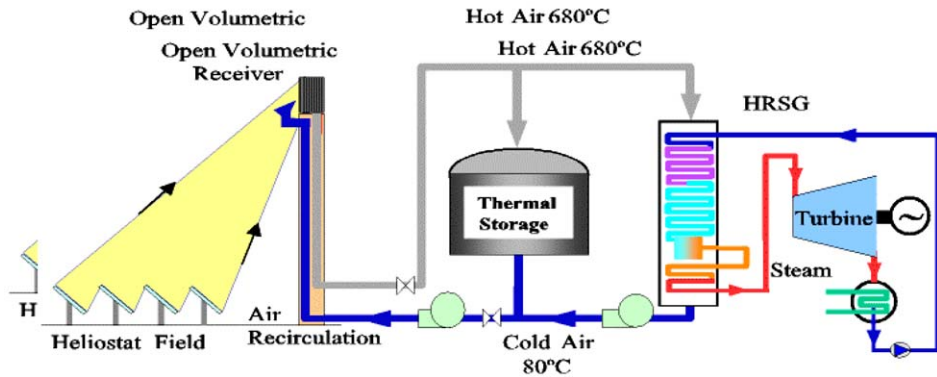


Fig. 2. Flow-chart of a power plant using tower technology.

Table 1

Optical, thermodynamic and resulting material requirements of candidate receiver materials

Optical/thermodynamic requirements	Material requirements
High absorption	Dark
Optical extinction	High porosity
Heat transfer surface	High cell density
High fluxes	Temperature resistance
Radial heat transport	Thermal conductivity
High permeability	3D-structure

flow direction of the air and thus to a more homogenous temperature distribution of the receiver. Air flow conditions in each channel are then dominated by the homogenized temperature of the receiver.

It could also be shown, that permeability properties of the porous material influence the homogeneity of the air flow significantly. This is mainly caused by the temperature-dependent viscosity of air, which increases with increasing temperature following a $T^{0.7}$ law. Consequently, having a uniform pressure difference over the length of a multi-channel array, the permeability in hot channels is lower. In some cases this may lead to a hot blocking of the channels and the formation of by-pass flows in cold channels. If the pressure loss ($\Delta p/L$) as a function of the fluid velocity (U_L) shows a linear shape according to the Darcy-law (L : length of the receiver, μ_{DYN} : dynamic viscosity of air, K : permeability)

$$\frac{\Delta p}{L} = \frac{\mu_{\text{DYN}}}{K} U_L$$

this tendency is significantly stronger compared to materials with a pressure loss following the Forchheimer-law describing a quadratic dependence of the pressure loss as a function of air flow velocity (K_1 : viscosity coefficient, K_2 : inertial coefficient,

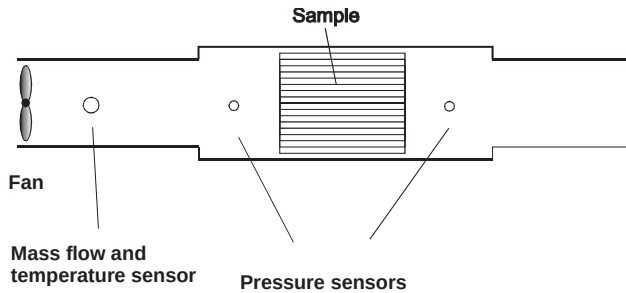


Fig. 3. Test set-up used for pressure loss measurements.

ρ_f : density of air):

$$\frac{\Delta p}{L} = \frac{\mu_{\text{DYN}}}{K_1} U_L + \frac{\rho_f}{K_2} U_L^2.$$

Generally, porous structures with straight channels show linear behavior, structures with changes in channel diameter show quadratic behavior.

Homogeneity of the air flow is a prerequisite for high efficiencies and the avoidance of local overheating, which may cause a destruction of the absorber. Past receiver concepts have been based on materials like metallic wire meshes, catalyst carriers with parallel channels for exhaust systems or ceramic foam materials, which have been optimized for other applications. In this paper two new approaches are reported, which are aimed at developing materials with the properties described above, giving special attention to a high ratio of K_1/K_2 : A porous silicon carbide material manufactured via the *Direct-Typing-Process* (called *typed material* in the further course of the text) and a fine celled silicon carbide foam (called *foam*). A detailed description of the materials themselves as well as the manufacturing technologies are described in the 3rd paragraph. The *foam* was developed and manufactured by the Fraunhofer IKTS¹ within a common project, the *typed material* was developed and manufactured by the company Bauer R&D within a subcontract according to the design of the authors. It was then delivered to DLR, where it was tested.

2. Experimental tests

This article presents results of permeability measurements and tests in concentrated solar radiation (solar furnace tests). *Permeability* measurements have been carried out with the simple set-up shown in Fig. 3. A fan generates an air flow through the porous sample. The pressure difference between the outlet and the inlet is measured. A mass flow and a temperature sensor allows the determination of the

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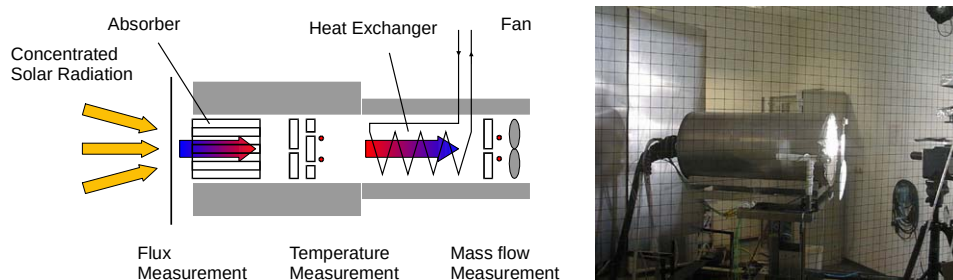


Fig. 4. Test bed used for open volumetric receiver efficiency measurement in concentrated solar radiation.

air flow velocity. Pressure loss as a function of air flow velocity can be measured in the range of 0–6 m/s. The accuracy of the measurements is mainly influenced by the accuracy of the pressure and temperature sensors. Thus the total accuracy is estimated to be $\pm 5\%$.

Solar furnace tests have been carried out in concentrated radiation using the DLR “Solar Furnace”, an installation consisting of a 25 m² Heliostat and a fixed concentrator [9]. This installation is capable to deliver a total power of up to 20 kW into the laboratory. In the focus of the furnace an isolated test-bed is used, in which receiver samples of cylindrical shape with diameters of up to 100 mm are placed (Fig. 4). For smaller diameters, an envelope out of insulating material is manufactured. A fan forces ambient air to flow through the sample followed by a air/water heat exchanger. The following quantities are monitored during the test (Tables 2 and 3): DNI², the material front (peak) temperature measured by a pyrometer, the air temperature at the outlet of the receiver, the air temperature at the outlet of the heat exchanger (HE air temp.), the total air mass flow, the heat exchanger water mass flow and the water temperature at the heat exchanger inlet and outlet. From these values the power of the air flow and the power of the water flow at the outlet of the heat exchanger can be calculated. Additionally to this permanently measured data, the optical flux density is measured, that penetrates through the circular aperture area (POA). This area corresponds to the diameter of the receiver sample. Secondly, the temperature distribution on the receiver front is measured. These two measurements are carried out at several times of the experiment (Tables 2 and 3). At these times efficiency has been calculated by (power water + power air)/POA with the POA value of this time and the 10 min averages of the other quantities. The POA values were taken to be representative for the last 10 min. This assumption was justified by the standard deviation of the DNI average to be less than 10 W/m². Optical flux measurements are performed with a camera/target method and temperature distribution measurements with an infrared camera. Details of the camera/target method have been previously published [9].

²Direct Normal Incidence (W/m²) of the solar radiation.

Table 2

Operational data of the efficiency test carried out with the 20 ppi foam receiver sample ($\varnothing$ 80 mm)

Time	DNI (W/m ²)	Power on aperture (POA) (kW)	Material front temp. (°C)	Air outlet Temp. (°C)	HE air temp. (°C)	Air mass flow (AMF) (kg/s)	HE water mass flow (l/min)	Power air (behind HE) (W)	Power water (W)	Efficiency (%)	POA/ AMF (kJ/kg)
11:07	812±5	2.88	606	455	41.5	0.0046	0.86	76.6	2119	76	626
11:28	817±3	3.694	715	577	44.8	0.0046	0.84	91.6	2687	75	803
11:55	858±8	4.417	707	545	42.4	0.0062	1.57	109.1	3434	80	710
12:35	860±2	4.8	737	593	44.0	0.0065	1.56	123.8	3876	83	743
13:05	875±3	5.352	803	655	45.4	0.0065	1.57	133.7	4304	83	823
13:19	876±3	5.322	803	663	46.0	0.0066	1.57	138.6	4400	85	810
13:45	873±3	4.79	725	614	45.8	0.0067	1.55	141.2	4188	90	710
14:15	863±3	4.336	713	570	45.3	0.0068	1.55	139.4	3931	94	636

All values but 'power on aperture' are 10 min averages of data taken every second. Standard deviations of the DNI values are added.

Table 3
Operational data of the efficiency test carried out with the 80/20 ppi foam receiver sample ($\varnothing$ 70 mm)

Time	DNI (W/m ²)	Power on aperture (POA) (kW)	Material front temp. (°C)	Air outlet temp. (°C)	HE air temp. (°C)	Air mass flow (AMF) (kg/s)	HE water mass flow (l/min)	Power air (behind HE) (W)	Power water (W)	Efficiency (%)	POA/AMF (kJ/kg)
10:19	769±4	2.774	628	431	37.7	0.0051	1.42	65.7	2239	83	541
11:01	788±1	3.552	774	576	41.5	0.0054	1.39	90.3	3176	92	652
11:37	800±1	4.043	843	651	43.4	0.0055	1.4	102.0	3633	92	733
12:04	804±2	3.734	800	618	43.5	0.0057	1.4	106.3	3562	98	655
13:03	807±3	3.681	794	606	43.8	0.0056	1.39	106.9	3463	97	652
13:31	798±5	4.142	867	672	44.9	0.0056	1.39	112.6	3821	95	736

The accuracy of the efficiency measurements is influenced by the accuracy of the temperature and the mass flow measurements as well as the optical flux measurements. The receiver sample and the tube, which connects receiver and heat exchanger is insulated with a 20 cm layer of different high insulating materials. However, there are heat losses from the tube through the insulation. They have been estimated to be lower than 2% at 800°C air temperature. The test bed itself has to be warmed up prior to steady state measurements. Continuous insolation is required for at least 30 min to keep the heat loss below the above mentioned value. The accuracy of the camera/target method for the POA measurements has been estimated to be $\pm 3.5\%$. A calculation, assuming a statistical error yields a total accuracy of the efficiency measurements of $\pm 4.8\%$.

3. Material technology

3.1. Ceramic foam receiver samples

Ceramic foams have been employed as volumetric receivers several times. The new approach reported here was to combine two foam materials of different cell density. The reason for this design is, that **a high cell density is only required in the extinction volume, the volume in which radiation is absorbed and the heat transfer material to air takes place**. The other part of the receiver may also consist of a **low cell density material with a higher permeability and a higher effective heat conductivity**. Thus, as a volumetric receiver test sample, a sandwich geometry as can be seen in Fig. 5 with the dimension $\varnothing 70 \times 30$ mm was supplied by the Fraunhofer IKTS within the common project SOLPOR. It consists of 20 ppi (= pores per inch) foam with a front layer of an 80 ppi foam. Additionally to this receiver sample a 20 ppi foam sample of dimensions $\varnothing 80 \times 30$ mm has been manufactured and tested as a basis for comparison.

Both materials consist of sintered silicon carbide (SSiC) with a heat conductivity of the walls of approximately 100 W/mK. They have been manufactured by a replication of polyurethane foams. The material was sintered at temperatures $> 2000^\circ\text{C}$ to achieve the required strength and high temperature resistivity of up to

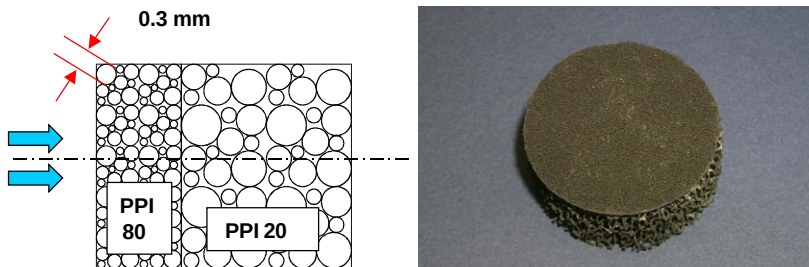


Fig. 5. Photograph and sketch of the 80/20 ppi silicon carbide foam receiver ($\varnothing 70$ mm).

1600°C. In general, the process allows the replication of foams with cell densities up to 100 ppi. Additionally, several infiltration techniques allow further reduction of wall porosity.

3.2. Receiver samples via the Direct-Typing-Process

Several receiver concepts of the last decade employed material developed for automotive catalysts. These materials, which consist of parallel channels and are manufactured by extrusion, showed beneficial properties with one exception: their linear permeability properties. As pointed out in the introduction this fact is disadvantageous with respect to flow stability. To overcome this, the Direct-Typing-Process was employed to create a 3D geometry, which is not possible with the extrusion process. Four cylindrical cellular receiver test samples with dimensions $\varnothing 100 \times 15 \text{ mm}$ have been manufactured by Bauer R&D within a subcontract. The geometry of the samples was designed by the authors. They consist of parallel channels with a square cross section and an open cell width of $1330 \mu\text{m}$. The wall thickness is $195 \mu\text{m}$. Additionally, and in contrast to extruded cellular monoliths, in each channel a plane of $270 \mu\text{m}$ thickness perpendicular to the channel direction was placed with a small hole of $150 \mu\text{m}$ diameter (Fig. 6). This additional plane changes

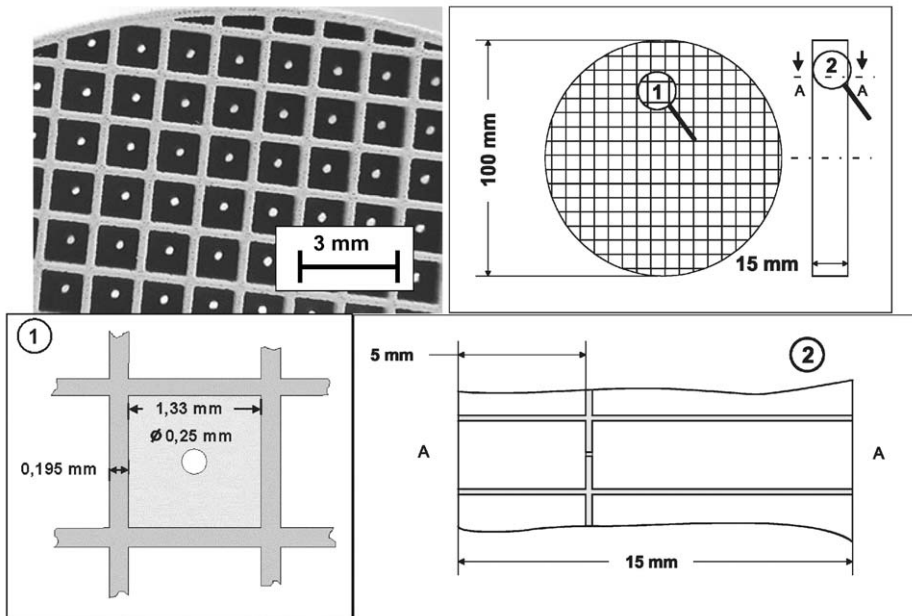


Fig. 6. The cellular silicon carbide receiver sample manufactured by Bauer R&D. Top left: Photograph of a front detail, top right: drawing of the receiver sample (top and lateral view), bottom left: detailed drawing of a single channel (front view), bottom right: drawing in section of a single channel (lateral view).

the fluid flow from pure Darcy flow to a more turbulent flow, which is an important prerequisite for an application as a volumetric receiver.

The *direct typing process* is a new method to create cellular bodies with pre-determined pore structures. Three dimensional objects (3D) can be manufactured in silicon carbide, a hard crystalline compound with notable abrasive and heat-resistant properties by means of the direct-typing modified screen-printing method. The 3D cellular bodies are built up from multiple thin layers in a way that allows much more constructive parameters than it is possible with conventional ceramic methods. Limitations in part size, shape, and elongation properties experienced with silicon carbide components made by extrusion or pressing are not an issue. And monolithic pieces retain thermal and other performance characteristics better than structures made by glueing several parts together. Further experiments by the process developer Bauer R&D Inc. suggest the possibility of integrating construction elements made of different materials within the monolithic structure.

For the process, a sieve-printable silicon carbide mass of low viscosity is necessary with which it is possible to print bodies with predetermined pore structures. Granular silicon carbide is mixed with a special organic additive prescription under atmospheric conditions. During an evacuation period the compound becomes a homogenized paste-like printing mass with long term stability. It hardens fast in thin layers and slowly in relatively thick ones. The diameter of the materials cellular pores can range from 0.25 to 5.00 mm. Using the patented production method, Bauer R&D has succeeded in processing this silicon carbide material into monolithic structures with wall thicknesses from 0.11 to 1.00 mm within one part and a base as large as 27.40–23.00 cm. A part height of 43.5 cm has been achieved with present equipment.

The final composition consists only of silicon carbide. Additives, used during the shaping procedure, are removed completely from the final product through heating, so that no sintering or other aids remain. The purity of the end product depends solely on the purity of the used granular silicon carbide. The resulting material properties of the structure are those of silicon carbide. The process uses no solvents and no waste gases, harmful to workers or the environment, are produced.

4. Results and discussion

The results of the *pressure loss measurements* are presented in Fig. 7. In this diagram the pressure loss is plotted as a function of the air velocity. The results obtained with the material manufactured via the *Direct-Typing-Process* (here denoted as “screenprinted material”) are compared with the results of a typical catalyst carrier with simple parallel channels without any artificial obstacles. The effect of the additional perforated plane in the *Direct-Typing-Process* material is significant. The pressure loss is higher but following a quadratic Forchheimer law. Also plotted in the same diagram are the results of the measurements with the sandwich-like silicon carbide foam, which also follows a Forchheimer-law as it is typical for foam materials.

The results of the tests with the 20/80 ppi sandwich foam receiver in concentrated solar radiation are presented in Fig. 8 along with the 20 ppi foam receiver for comparison. In this case measurement time and insolation conditions were sufficient to determine detailed efficiency information from the tests. It can be seen, that the efficiency of the sandwich material is approximately 10% higher compared to the 20 ppi receiver. This is mainly caused by the larger heat transfer surface available, which is $5400 \text{ m}^2/\text{m}^3$ for the 80 ppi foam compared to $1100 \text{ m}^2/\text{m}^3$ for the 20 ppi foam. The larger heat transfer surface causes an improved cooling of the front surface leading to lower radiative losses. As a basis for an additional comparison, results of previous studies are shown in Fig. 9 [7]. The metallic fiber mesh material was employed within the PHOEBUS project [3], and the SiC catalyst carrier material is currently approved on a 3 MW scale within the European project SOLAIR [11].

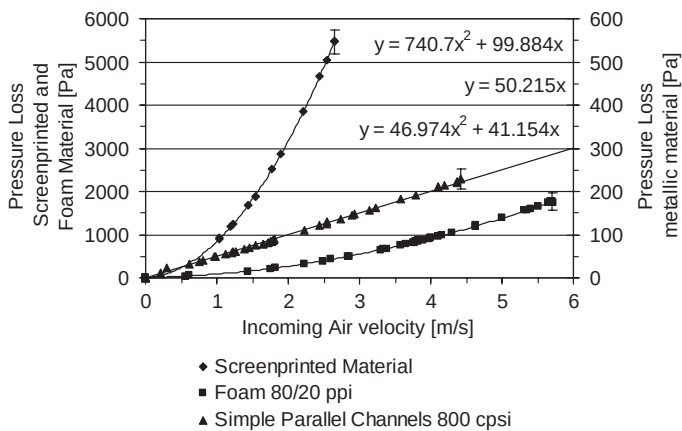


Fig. 7. Results of the pressure loss measurements comparing the ceramic foam material, the *Direct-Typing-Process* material and a simple parallel channel catalyst carrier material.

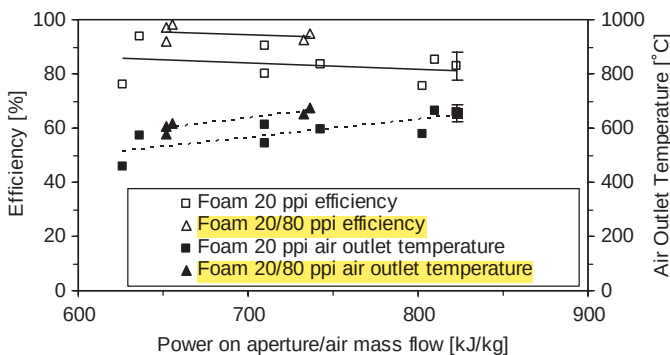


Fig. 8. Results of the efficiency measurements of the sandwich-foam-material compared to a simple foam material.

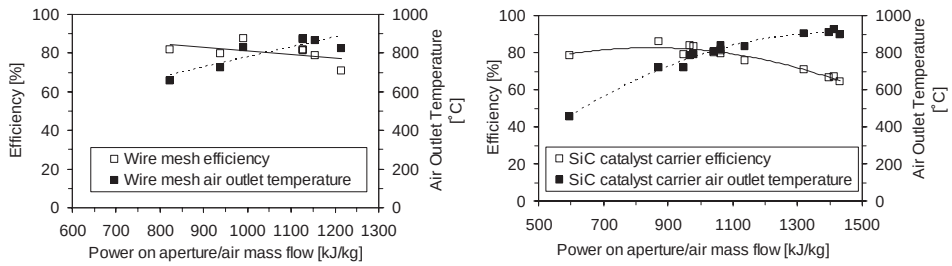


Fig. 9. Efficiency properties of a wire mesh and a silicon carbide catalyst carrier material, both having been tested in previous studies as a basis for comparison [7].

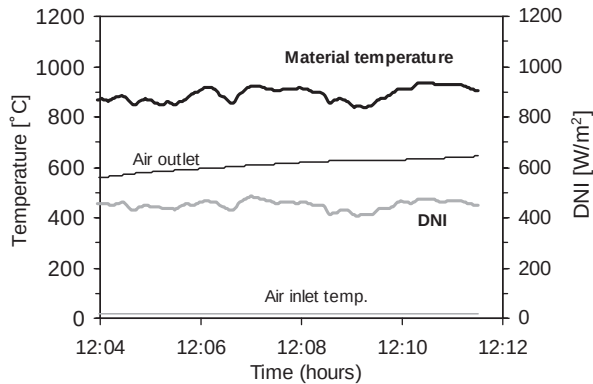


Fig. 10. Temperature–Time plots during the test of the *Direct-Typing-Process* material in concentrated solar radiation.

The results of the tests with the *printed material in concentrated solar radiation* are presented in Figs. 10 and 11. An excerpt of the temperature vs. time plot of the test of the *Direct-Typing-Process* material can be seen in Fig. 10. Air inlet and outlet, as well as material temperature measured with a pyrometer at the front of the sample can be seen. Additionally, solar direct normal incidence power is plotted. Due to the short time of the measurement and due to the strong fluctuation of the incident power caused by weather conditions the receiver efficiency could only be roughly estimated by averaging. It was 80% at a power of 2.3 kW on the aperture. However, the average air outlet temperature was up to 650°C. Compared to the a metallic parallel channel structure with linear pressure loss characteristics, this is a significant improvement. Efficiency measurement results of the metallic parallel channel structure have been previously published [10] and are added for comparison in Fig. 12. Both materials have comparable geometric channel dimensions and optical properties, thus, pressure loss as a function of fluid velocity characteristics changing from linear to quadratic cause this difference. The corresponding front temperature

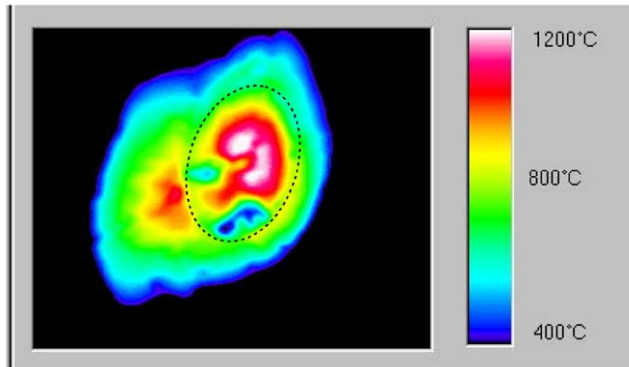


Fig. 11. Front temperature distribution on the test sample of the *Direct-Typing-Process* material during the test in concentrated solar radiation. The receiver aperture is marked by the dotted line.

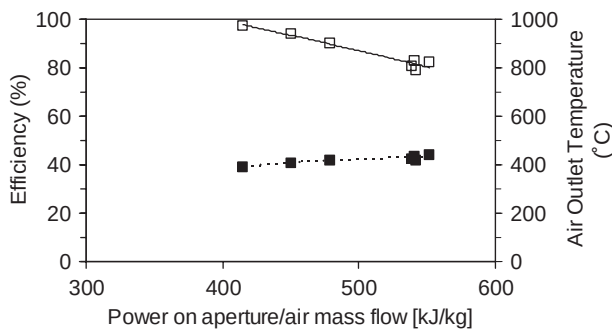


Fig. 12. Efficiency properties of an 800 cpsi (channels per square inch) metallic catalyst carrier material with a geometry similar to the printed material from a previous study for comparison [10].

distribution is shown in Fig. 11. It can be seen that peak temperatures of about 1200°C occur. Note that the incoming concentrated flux is Gaussian like ranging from 0.5 MW/m^2 in the middle to 0.15 MW/m^2 at the outer areas of the aperture.

5. Conclusions

Two novel porous materials for an application as a volumetric receiver have been presented and tested. First, a double layer silicon carbide foam showed improved performance in an experiment with concentrated solar radiation compared to a single layer material, which has already been applied before. Secondly, the *direct typing process* has been employed to show the feasibility of manufacturing a porous ceramic structure with predefined permeability properties. Material samples, meeting

the general requirements for an application as a volumetric receiver have been manufactured and pre-tests in concentrated solar radiation have been performed with promising results. Both technologies, the direct typing process and the ceramic foam technology are a future option for the production of materials with properties required for the use as a high-performance solar receiver. Encouraged by the results reported, tests of receiver modules for the 200 kW solar tower test facility at the Plataforma Solar in Almería, Spain are planned. Special attention will be paid on long-term high-temperature strength and oxidation resistance.

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